

STATISTICS FOR DATA ANALYTICS

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Module Leader: Dr. Muhammad Alli

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Student name: Van Ly Nguyen



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INTRODUCTION

Car Details dataset is taken from Kaggle (<https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho>) and it contains 8 columns with 4340 records. This dataset describes the details of cars and its various features using five - categorical variables and three - continuous variables. The analysis of car selling prices provides valuable insights into market trends, pricing strategies, and vehicle depreciation over time. This study examines a dataset of used car details from CarDekho to explore statistical properties, visualizations, and the presence of outliers in the dataset.

DATA OVERVIEW

The dataset was loaded into R using `cars <- read.csv ()` and the first few records were examined using `head (cars)`. The dataset contains several key variables, including:

- **name:** Car model
- **year:** Year of manufacture
- **selling_price:** The price at which the car was sold
- **km_driven:** Distance covered by the vehicle
- **fuel:** Fuel type (Petrol, Diesel, CNG, LPG, Electric)
- **seller_type:** Whether the seller is an individual or dealer
- **transmission:** Manual or Automatic
- **owner:** Number of previous owners

Question 1

a) Describe the dataset using appropriate plots/curves/charts

Code:

```
cars = read.csv(file.choose()) # choose the dataset file
head(cars)
```

Result:

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner
1	Maruti 800 AC	2007	60000	70000	Petrol	Individual	Manual	First Owner
2	Maruti Wagon R LXI Minor	2007	135000	50000	Petrol	Individual	Manual	First Owner
3	Hyundai Verna 1.6 SX	2012	600000	100000	Diesel	Individual	Manual	First Owner
4	Datsun RediGO T Option	2017	250000	46000	Petrol	Individual	Manual	First Owner
5	Honda Amaze VX i-DTEC	2014	450000	141000	Diesel	Individual	Manual	Second Owner
6	Maruti Alto LX BSIII	2007	140000	125000	Petrol	Individual	Manual	First Owner

Various visualization techniques were used to explore the dataset:

Here I choose selling_price as the attribute. Because,

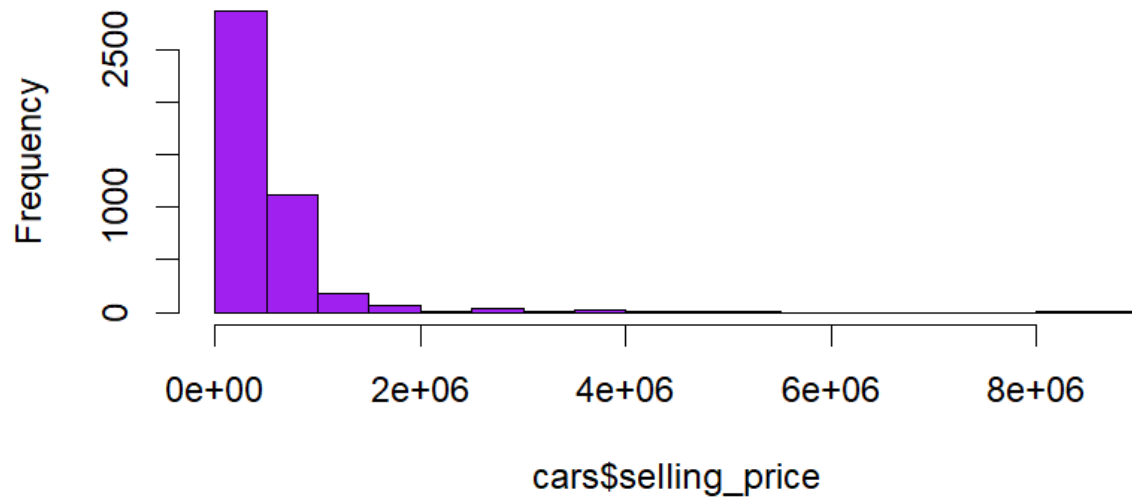
- **Primary Business Metric:** One important financial indicator that establishes a car's worth in the marketplace is its selling price. In predictive modelling, the dependent variable is frequently the target variable and aids in estimating automobile prices by taking into account other factors.
- **Market insights:** Examining selling prices provides information on depreciation rates, pricing patterns, and the impact of various factors on car valuation.
- **Buyer and Seller Decision-Making:** To make well-informed choices, buyers and sellers consult selling price analysis.
- **Comparison Across Categories:** To identify price variances and patterns, the selling price can be examined across a number of categories, including fuel type, transmission, owner history, etc.

Histogram:

Code: `hist(cars$selling_price, col='purple')`

Result:

Histogram of cars\$selling_price

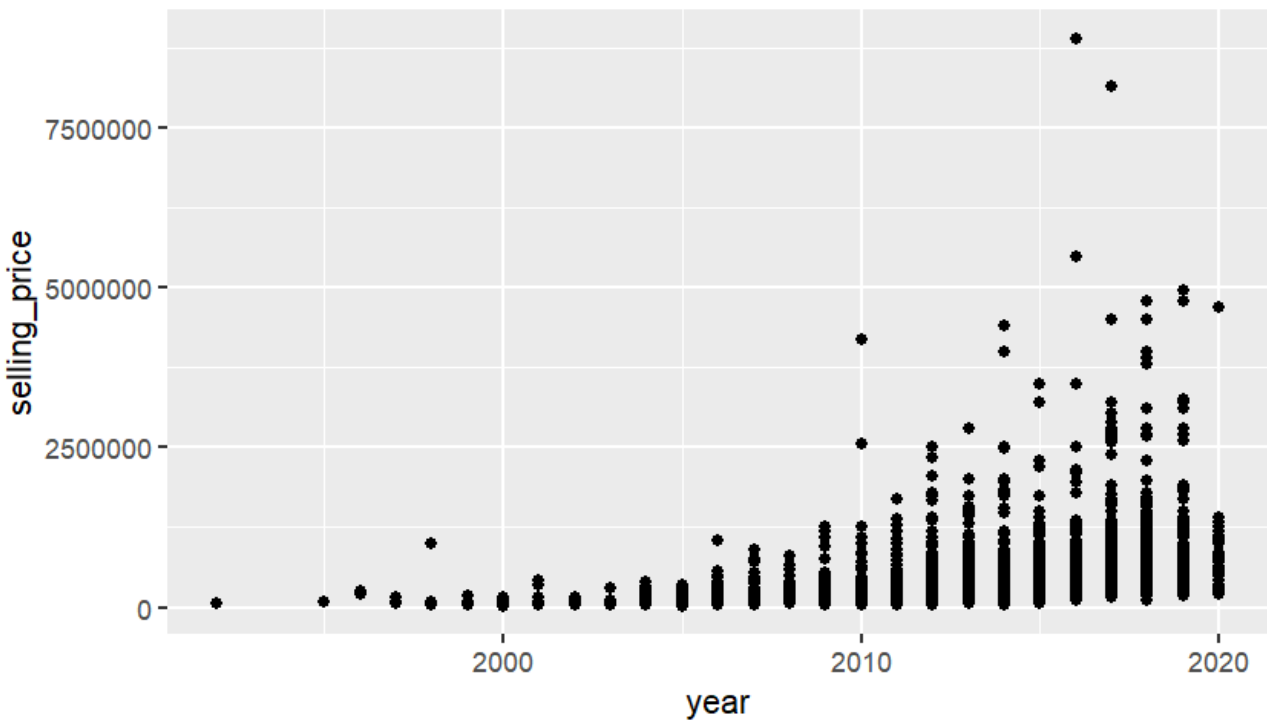


The dataset's distribution of car selling prices is displayed via the histogram. There are fewer cars at higher costs and more cars at lower price ranges, suggesting a distribution that is biased to the right.

Scatterplot:

Code:

```
install.packages("ggplot2")
library(ggplot2)
ggplot(data = cars, aes(x = year, y = selling_price)) +
  geom_point()
```

Result:

The scatter plot displays the correlation between car sales price and the year of manufacturing.

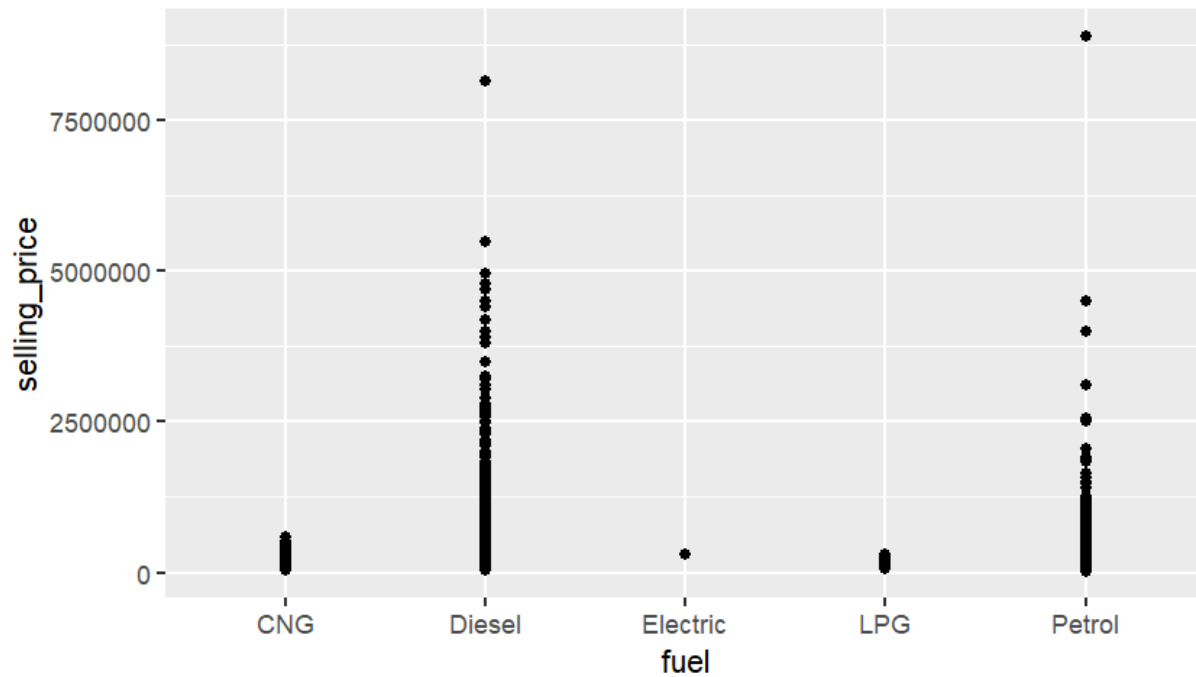
Important findings:

Older models typically have lower selling prices, whereas newer models (those manufactured in more recent years) typically have higher pricing. There is a downward trend: the car's value drops with increasing age. There are certain exceptions, where older cars sell for comparatively high prices, maybe as a result of their luxury or excellent upkeep.

Code:

```
ggplot(data = cars, aes(x = fuel, y = selling_price)) +  
geom_point()
```

Result:



Distribution of Prices by Fuel Type:

- The dataset is dominated by Petrol and Diesel vehicles.
- Generally speaking, diesel vehicles are more expensive to buy than Petrol vehicles.
- Prices for CNG and LPG vehicles are comparatively lower, perhaps as a result of decreased demand.
- Although they are less common, electric automobiles might vary more in price.

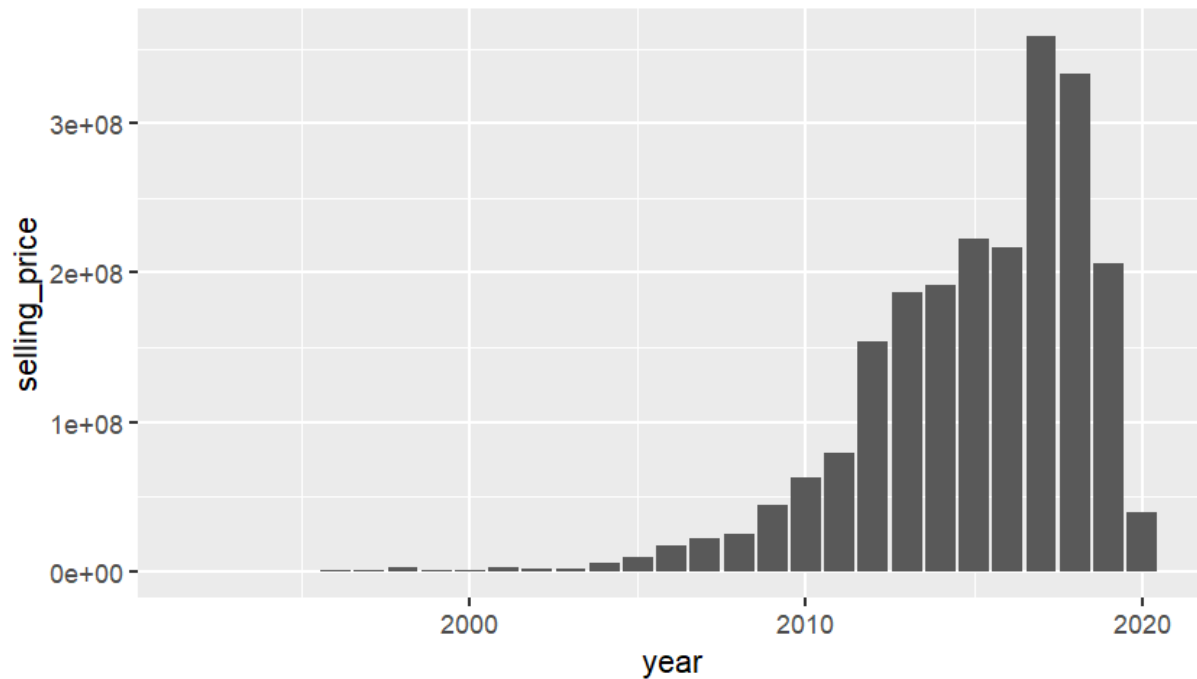
Trends and Observations:

- Due to their superior fuel efficiency for long-distance driving, diesel cars typically have greater resale values.
- Prices for gasoline-powered vehicles vary widely, suggesting a mix of high-end and low-cost models.
- Cars that run on CNG and LPG serve a specialized market and might not be as popular.

Bar Plot:

Code: `ggplot(data = cars, aes(x=year, y=selling_price)) +
 geom_bar(stat="identity")`

Result:



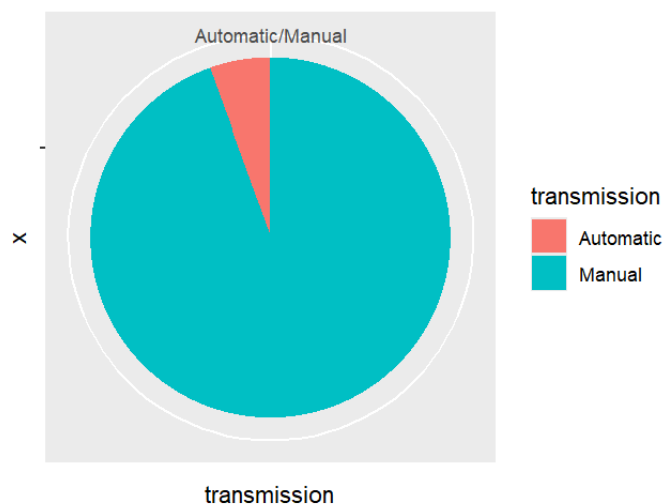
Higher total selling prices in recent years: Newer and more costly cars make up a larger portion of the total cost. Reduced total selling prices for older years: The resale value of older cars is typically lower. Potential increases in specific years: Variability may arise from higher car prices or more transactions in some years.

Pie Chart:

Code:

```
ggplot(cars, aes(x="", y=transmission, fill=transmission)) +  
geom_bar(stat="identity", width=1) + coord_polar("y", start=0)
```

Result:



Transmission Type: Automatic vs. Manual:

Manual transmissions are seen in the vast majority of the cars in the dataset. Automobiles with automatic transmissions make up a lesser percentage of the sample.

Potential Causes of Distribution:

Manual cars are more common in the used car market, as they are generally more affordable and fuel-efficient. Although they are becoming more and more popular, automatic cars have a lower share of the resale market because they are typically more expensive.

Market Implications:

If you're looking to buy a used car, you'll likely find more manual transmission options. Sellers of automatic cars may find a more niche but possibly higher-paying customer base.

b) Consider one of continuous attributes, and compute central and variational measures.

CENTRAL MEASURES

Mean is the average of all values. Here we are calculating the average of selling_price

MEAN SELLING PRICE:

Code:

```
mean_Price=mean(cars$selling_price)
mean_Price
```

Result: [1] 504127.3

MEDIAN SELLING PRICE:

Median is the middle value in the column selling_price when it is sorted order.

Code:

```
median_Price=median(cars$selling_price)
median_Price
```

Result: [1] 350000

MODE SELLING PRICE:

Determined by finding the most frequent price, which indicated that some price points were more common than others.

Code:

```
Price_table =table(cars$selling_price)
mode_Price=names(Price_table)[which(Price_table==max(Price_table
))]
mode_Price
```

Result: [1] "300000"

The distribution is right-skewed, implying that a small number of expensive cars are driving up the mean, which is higher than the median. Since 50% of the cars are priced below ₹350,000 and 50% above, the median denotes the middle value. According to the mode, the most frequently listed selling price is ₹300,000.

VARIATIONAL MEASURES

RANGE:

Range is the difference between the highest and lowest value in the column.

Code:

```
range_Price = max(cars$selling_price)-min(cars$selling_price)
range_Price
```

Result: [1] 8880000

INTERQUARTILE RANGE (IQR):

IQR is the range of values located in the midpoint of the distribution of values. The 25th percentile, or Q1 (First Quartile), indicates that 25% of the data. The third quarter, or Q3, is the 75th percentile, which indicates that 75% of the data. The difference between Q3 and Q1, which measures the middle 50% of the data.

Code:

```
Q=quantile(cars$selling_price)
Q
```

Result:

0%	25%	50%	75%	100%
20000.0	208749.8	350000.0	600000.0	8900000.0

Code:

```
Q1 <- quantile(cars$selling_price, 0.25)
print(Q1)
```

Result: 25%
208749.8

Code:

```
Q3 <- quantile(cars$selling_price, 0.75)
print(Q3)
```

Result: 75%
6e+05

Code:

```
IQR = Q3-Q1
IQR
```

Result: 75%

391250.2

VARIANCE:

`var(cars$selling_price)`, providing another measure of data dispersion. Measures how far each data point is from the mean (average).

Code:

```
variance_Price = var(cars$selling_price)
variance_Price
```

Result: [1] 334718640088

STANDARD DEVIATION:

It shows how much prices deviate from the mean.

Code:

```
SD_Price = sd(cars$selling_price)
SD_Price
```

Result: [1] 578548.7

COEFFICIENT OF VARIATION:

Measures the relative variability as a percentage of the mean. Useful for comparing variability between datasets with different units

Code:

```
cv <- (SD_Price / mean_Price) * 100
cv
```

Result: [1] 114.7624

MEAN ABSOLUTE DEVIATION:

The average absolute difference between each data point and the mean.

Code:

```
mad_value <- mad(cars$selling_price, center =
mean(cars$selling_price), constant = 1)
mad_value
```

Result: [1] 246372.7

Conclusion: The selling prices range from ₹ 8880000, indicating huge price discrepancy. The interquartile range (IQR) is ₹391,250.2, representing the middle 50% of prices. The standard deviation of ₹578,548.74 indicates substantial variety in car costs. The coefficient of variation (CV) is 114.76%, indicating significant relative fluctuation. The mean absolute deviation (MAD) is ₹246372.7, which is the departure from the mean. These measurements corroborate a widely spread dataset, impacted by extreme values and price volatility.

c) For a particular variable of the dataset, use Chebyshev's rule, and propose one-sigma interval. Based on your proposed interval, specify the outliers if any.

The lower and upper ranges for locating possible outliers were determined using Chebyshev's theorem:

Lower Range: $\text{mean} - k * \text{standard deviation}$

Upper Range: $\text{mean} + k * \text{standard deviation}$

The existence of extreme values in the dataset can be verified by looking at the number of outliers (`sum(outliers)`).

Code:

```
k = 1
lowerRange = mean_Price - k * SD_Price
upperRange = mean_Price + k * SD_Price
round(lowerRange, 2)
```

Result: [1] -74421.42

Code:

```
round(upperRange, 2)
```

Result: [1] 1082676

Code:

```
outliers = cars$Selling_Price < lowerRange | cars$Selling_Price >
upperRange
sum(outliers)
```

Result: [1] 0

Here, we calculate the One Sigma Interval using **k=1**

```
lowerRange = mean_Price - k * SD_Price
upperRange = mean_Price + k * SD_Price
```

Therefore, the One Sigma Interval for selling_price is

Lower Range= -74421.42

Upper Range= 1082671

Normal values fall within a one-sigma interval (mean standard deviation) of -74,421 to 1,082,676. Any value not inside this range is considered an anomaly. Here `sum(outliers)=0`, based on this, we can say that all values in the selling_price falls within the calculated lower range and upper range. According to Chebyshev's rule or one-sigma rule there are no outliers in this dataset.

d) Explain how the box-plot technique can be used to detect outliers. Apply this technique for one attribute of the dataset

A box plot is a useful graphical approach for identifying outliers in a dataset. It visualises a numerical variable's distribution by displaying its median, quartiles, and extreme values (minimum and maximum). Outliers can be found by calculating the distance between the quartiles (interquartile range, or IQR) and the distribution's extremes. An outlier is an observation that is more than 1.5 times the interquartile range below or above the first quartile. In other words, an outlier is an observation that is outside the range defined by the box plot's whiskers. Any data point below the lower bound or above the upper bound is classified as an outlier. Consider the `selling_price` attribute when applying this strategy to find outliers. We can make a box plot for this variable using R, as follows:

A boxplot was created (`boxplot(cars$selling_price)`) to identify statistical outliers.

STEPS FOR DETECTING BOX PLOT TECHNIQUE

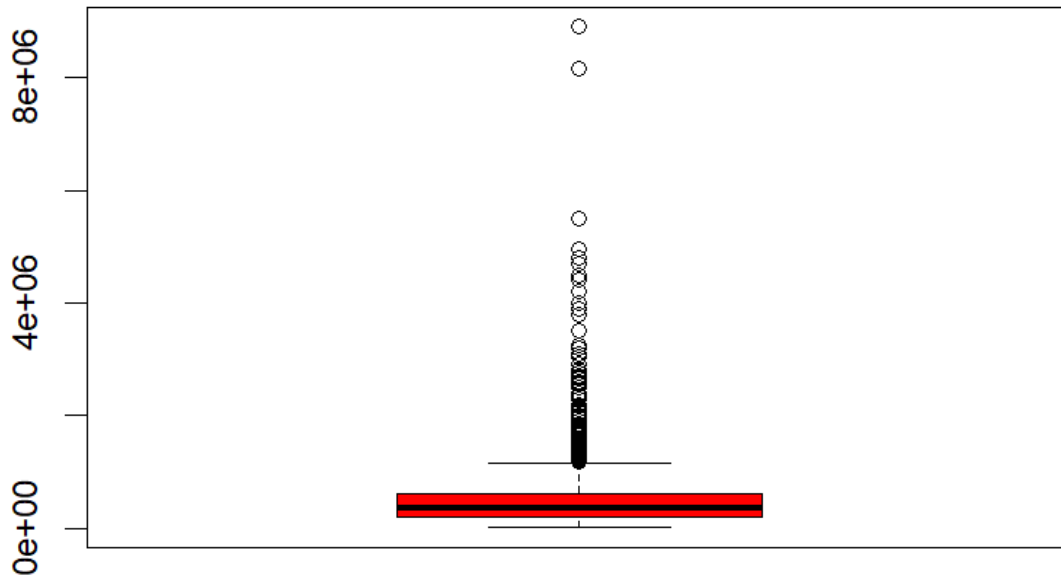
Step 1 : Create a box plot of `selling_price`

Code:

```
boxplot(cars$selling_price, main = "Box Plot of selling_price",  
, col="red")
```

Result:

Box Plot of selling_price



The above graph is a box plot of the `selling_price` attribute, which shows the data distribution and the location of any outliers. If there are any outliers, they will be displayed as solitary points outside the whiskers of the box plot. This box plot can help you detect potential outliers and assess if they are real data points or data entry errors. If they are genuine data points, we must decide whether to keep or discard them from the analysis, depending on the research topic and the impact of the outliers on the results.

Step 2: Calculate Quartiles and Interquartile Range (IQR)

Code:

```
q = quantile(cars$selling_price, c(0.25, 0.75))
iqr = q[2] - q[1]
print(iqr)
```

Result: 75%
 391250.2

Step 3: Define lower and upper bounds

```
upperOutliers = q[2] + 1.5 * iqr
print(upperOutliers)
```

Result: 75%
1186875

```
lowerOutliers = q[1] - 1.5 * iqr  
print(lowerOutliers)
```

Result: 25%
-378125.6

Step 4: Identify outliers

Code:

```
boxplotOutliers = cars$selling_price > upperOutliers |  
cars$selling_price < lowerOutliers  
sum(boxplotOutliers)
```

Result: [1] 271

The number of extreme outliers was counted (sum(boxplotOutliers)), revealing significant deviations.

Code:

```
print(boxplotOutliers)
```

Result:

```
> print(boxplotOutliers)  
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[24] FALSE FALSE TRUE FALSE TRUE FALSE TRUE TRUE FALSE TRUE FALSE FALSE TRUE TRUE FALSE FALSE TRUE TRUE FALSE FALSE TRUE TRUE FALSE  
[47] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[70] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE  
[93] FALSE FALSE TRUE FALSE TRUE FALSE FALSE FALSE FALSE TRUE TRUE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[116] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[139] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[162] FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[185] FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE  
[208] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE  
[231] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[254] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[277] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE  
[300] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE TRUE  
[323] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[346] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[369] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[392] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE  
[415] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE  
[438] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[461] FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[484] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[507] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[530] FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE  
[553] FALSE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE
```

- Total Outliers Identified: 271 data points
- Percentage of Outliers: 6.24% of the dataset

Code:

```
outlier_cars <- cars[boxplotOutliers, c("name", "selling_price")]  
print(outlier_cars)
```

Result:

```
> print(outlier_cars)
```

	name	selling_price
13	Toyota Corolla Altis 1.8 VL CVT	1650000
26	Toyota Corolla Altis 1.8 VL CVT	1650000
28	Hyundai Venue SX Opt Diesel	1195000
30	Jaguar XF 2.2 Litre Luxury	1964999
31	Mercedes-Benz New C-Class 220 CDI AT	1425000
33	Audi Q5 2.0 TDI	1190000
36	Audi A6 2.0 TDI Design Edition	1735000
37	Mercedes-Benz New C-Class C 220 CDI Avantgarde	1375000
40	Toyota Innova 2.5 G (Diesel) 7 Seater	1300000
41	Jeep Compass 1.4 Sport Plus BSIV	1400000
44	BMW 3 Series 320d Sport Line	1550000
45	Audi Q7 3.5 TDI Quattro Premium	1250000
90	Mercedes-Benz S-Class S 350d Connoisseurs Edition	8150000
95	Audi A6 2.0 TDI Premium Plus	1470000
97	Audi A8 4.2 TDI	2800000
102	Mercedes-Benz E-Class Exclusive E 200 BSIV	4500000
103	BMW X1 sDrive 20d xLine	2750000
104	Volvo V40 D3 R Design	1975000
106	BMW 7 Series 730Ld	2500000
164	Jaguar XJ 5.0 L V8 Supercharged	2550000
192	Mercedes-Benz New C-Class 220 CDI AT	1500000
205	BMW 5 Series 520d Luxury Line	2900000
219	Mercedes-Benz New C-Class C 220 CDI BE Avantgare	2000000
228	Mahindra Scorpio S11 BSIV	1500000
290	Mahindra XUV500 W11 Option AWD	1400000
292	Mahindra Marazzo M8 8Str	1300000
319	BMW 3 Series 320d Sport	2000000

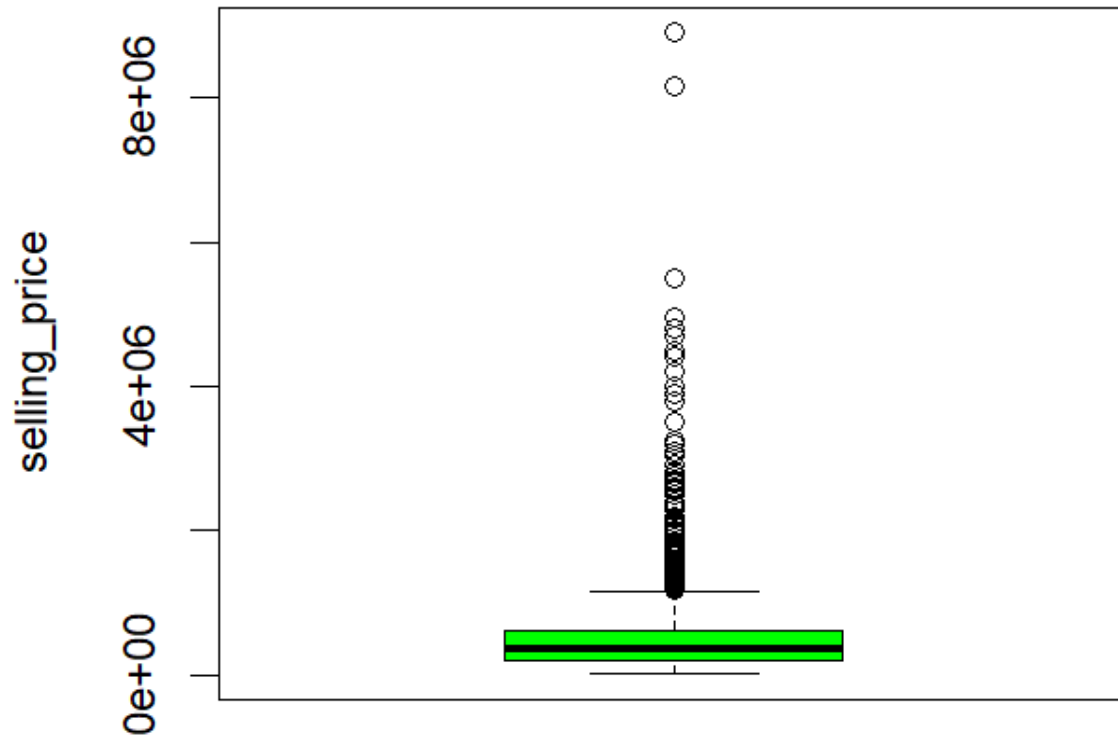
Step 5: Highlight outliers on the boxplot

Code:

```
boxplot(cars$selling_price, main="Boxplot with Outliers",
ylab="selling_rice", col="green")
points(which(cars$selling_price %in% outliers), outliers,
col="yellow", pch=16)
```

Result:

Boxplot with Outliers



Question 2

- a) **Select four variables of the dataset, and propose an appropriate probability model to quantify uncertainty of each variable.**

We are examining the following four variables to implement various probability models that measure the uncertainty associated with each variable:

1. Selling_Price
2. Fuel
3. Transmission
4. Kms_Driven

The following are the recommended appropriate probability models for the aforementioned variables:

1. Log-Normal Distribution
2. Multinomial Distribution
3. Bernoulli Distribution
4. Gamma Distribution

Let us examine in detail the variables and their proposed probability models:

1. The 'selling_price' column comprises continuous numerical data indicative of the vehicle's pricing. Due to the continuous nature of the data, we can employ a 'Log-Normal Distribution' for modeling purposes. By utilizing the sample mean and sample standard deviation of the Selling Price column, we may approximate the parameters of the log normal distribution (mean and standard deviation).
2. The 'fuel' column has categorical data indicating the type of fuel utilized by the vehicle (petrol, diesel, or CNG). A 'Multinomial Distribution' quantifies the probability of a singular occurrence among multiple potential outcomes and can be employed to simulate the likelihood of a vehicle utilizing petrol, diesel, or CNG as fuel. By utilizing the proportion of vehicles in the dataset that utilize each fuel type, we can estimate the parameters of the multinomial distribution, namely the probabilities associated with each fuel type employed by the vehicles.
3. The 'transmission' column has categorical data denoting whether the vehicle features a manual or automated transmission. A Bernoulli distribution, which quantifies the probability of a singular binary outcome, can be employed to describe this scenario; specifically, the likelihood of the car possessing an automatic transmission. By utilizing the ratio of vehicles in the dataset equipped with automatic transmissions, we can estimate the parameter of the Bernoulli distribution, which represents the probability of a vehicle possessing an automatic gearbox.
4. The 'Kms_Driven' column comprises continuous numerical data indicating the total kilometers driven by the vehicle. This statistic is consistently positive and often right-skewed, with the majority of vehicles exhibiting modest mileage and a minority displaying exceptionally high mileage. The Gamma Distribution serves as a suitable probability model for assessing uncertainty regarding this variable.

b) For each model in part (a), estimate the parameters of model.

1. For variable Selling_Price using Log-Normal Distribution

A variable XX follows a Log-Normal distribution if its logarithm ($\ln(X)$) is Normally distributed. It is characterized by:

- **Meanlog** (μ): The mean of the natural logarithm of X .
- **Sdlog** (σ): The standard deviation of the natural logarithm of X .

Parameters:

- μ : Mean of ($\ln(X)$).
- σ : Standard deviation of ($\ln(X)$).

Code:

```
log_selling_price <- log(cars$selling_price)
```

```
meanlog <- mean(log_selling_price, na.rm = TRUE)
meanlog
```

Result:

```
[1] 12.76418
```

Firstly, I transform the selling price into its logarithmic form. The purpose of this transformation is to stabilize variance and normalize the data. By applying a natural logarithm, we can better visualize relationships and perform parametric tests that assume normality.

The mean of the log-transformed selling prices is 12.76418. This value represents the average log-transformed price, which we can interpret on a logarithmic scale. This can be used as a baseline to understand the typical selling price in the transformed space.

Code:

```
sdlog <- sd(log_selling_price, na.rm = TRUE)
sdlog
```

Result:

```
[1] 0.8392654
```

The standard deviation of **0.839** indicates the variability of the log-transformed selling prices. A lower standard deviation suggests that the prices, when transformed, are relatively less spread out, providing some insight into the consistency of the selling

Code:

```
# Create a histogram of selling_price

hist(cars$selling_price, breaks = 30, probability = TRUE, main =
"Log-Normal Fit - Selling Price", xlab = "Selling Price")

# Define x as a sequence of values across the range of
selling_price
```

```
x <- seq(min(cars$selling_price, na.rm = TRUE),
max(cars$selling_price, na.rm = TRUE), length = 100)

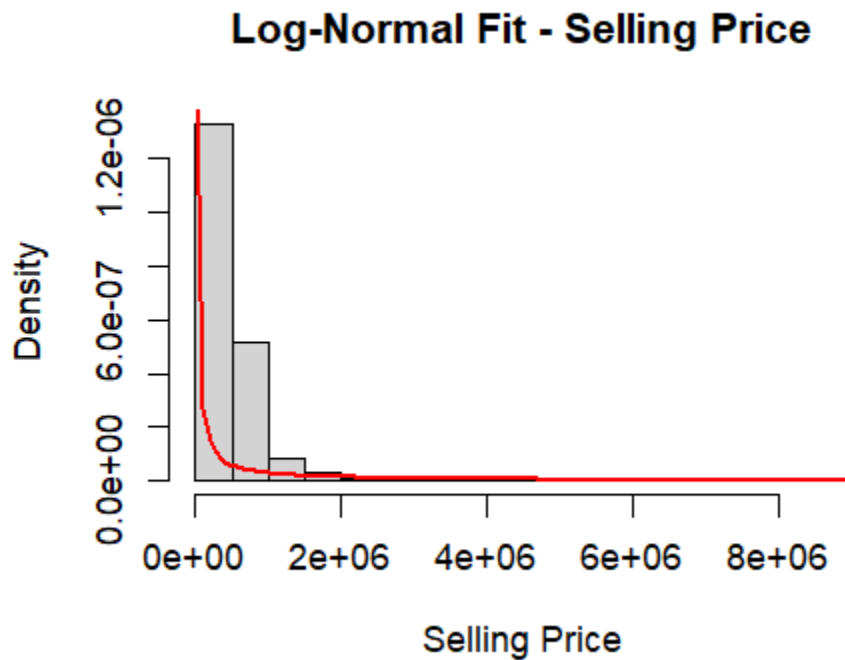
# Calculate the Log-Normal PDF

log_normal_pdf <- dlnorm(x, meanlog = meanlog, sdlog = sdlog)

# Overlay the Log-Normal curve on the histogram

lines(x, log_normal_pdf, col = "red", lwd = 2)
```

Result:



The histogram provides a graphical representation of the distribution of the original selling prices. As expected, the distribution appears right-skewed, with a long tail on the higher end.

The red line represents the theoretical log-normal distribution based on the calculated parameters. If the histogram closely matches the red line, it would suggest that the selling prices follow a log-normal distribution.

2. For variable Fuel using Multinomial Distribution

Parameters of multinomial distribution are defined by the number of trials n and a set of probabilities $p_1, p_2, \dots, p_k$ for each category.

Code:

```
X=cars$fuel  
t=table(X); t
```

Result:

```
X  
   CNG   Diesel Electric   LPG   Petrol  
   40    2153         1    23    2123
```

In this code, I created a **frequency table** of the `fuel` column from the `cars` dataset using the `table()` function. The output shows the count of each fuel type:

- **CNG**: 40 occurrences
- **Diesel**: 2,153 occurrences
- **Electric**: 1 occurrence
- **LPG**: 23 occurrences
- **Petrol**: 2,123 occurrences

From this, it is clear that **Diesel** and **Petrol** are the most common fuel types in the dataset, while **Electric**, **LPG**, and **CNG** are much less frequent.

Code:

```
> mode=names(t)[which(t==max(t))]; print(mode)
```

Result:

```
[1] "Diesel"
```

The `which(t == max(t))` function returns the index of the highest value in the frequency table `t`. The `names(t)` function is then used to extract the corresponding fuel type. As expected, the most frequent fuel type is **Diesel**, which occurs 2,153 times

Code:

```
p=t/sum(t);p
```

Result:

```
X
      CNG      Diesel      Electric      LPG      Petrol
0.0092165899 0.4960829493 0.0002304147 0.0052995392 0.4891705069
```

Here, I calculated the proportion of each fuel type by dividing the frequency of each fuel type (`t`) by the total number of fuel entries (`sum(t)`). The resulting proportions are as follows:

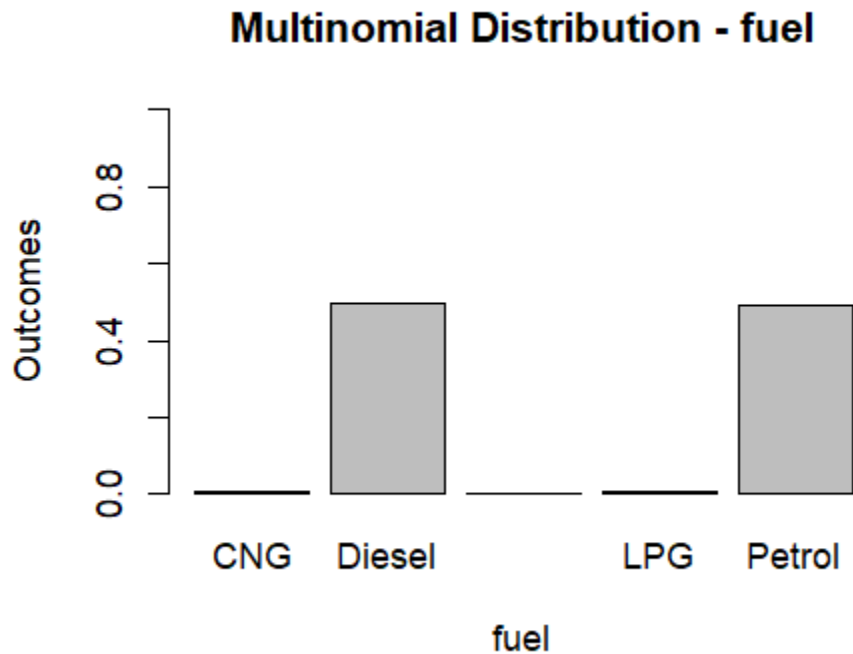
- **CNG**: 0.0092 (0.92% of the total dataset)
- **Diesel**: 0.4961 (49.61% of the total dataset)
- **Electric**: 0.00023 (0.023% of the total dataset)
- **LPG**: 0.0053 (0.53% of the total dataset)
- **Petrol**: 0.4892 (48.92% of the total dataset)

From this, we see that **Diesel** and **Petrol** account for nearly all of the dataset, with Diesel being slightly more frequent than Petrol. In contrast, **Electric**, **LPG**, and **CNG** are quite rare.

Code:

```
barplot(p, main = "Multinomial Distribution - fuel", xlab =
"fuel", ylab = "Outcomes", ylim = c(0, 1))
```

Result:



Finally, I created a **bar plot** to visualize the proportions of each fuel type. This code generates a bar plot with the fuel types on the x-axis and their corresponding proportions on the y-axis.

The plot has the following features:

- **Title:** "Multinomial Distribution - fuel"
- **x-axis label:** "fuel"
- **y-axis label:** "Outcomes"
- **y-axis range:** from 0 to 1, representing proportions.

From the bar plot, we would expect **Diesel** and **Petrol** to have the tallest bars, confirming their dominance in the dataset, while the bars for **CNG**, **Electric**, and **LPG** would be much smaller.

3. For variable Transmission using Bernoulli Distribution

The Bernoulli distribution has a single parameter p , which is the probability of success in a single trial.

Code:

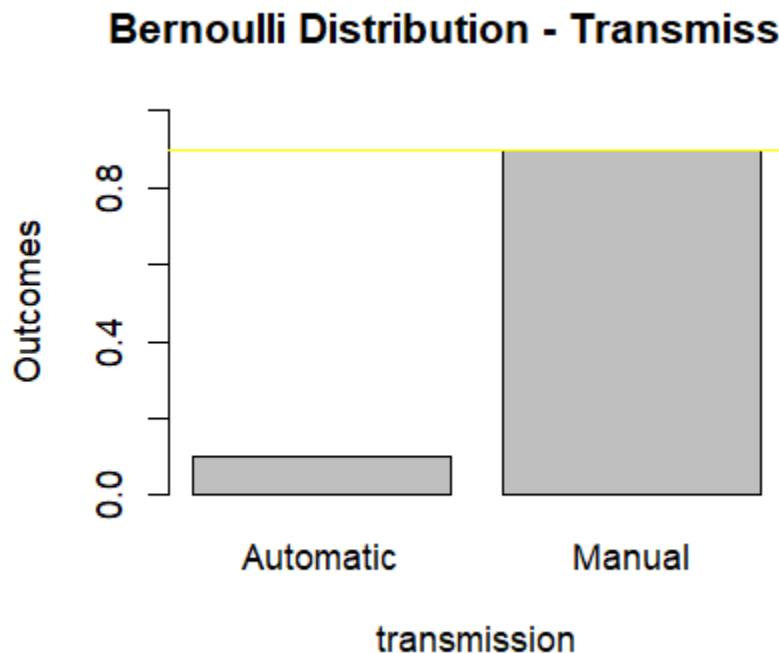
```
transmission = cars$transmission
transmission
transmission_prop = table(transmission) / nrow(data)
transmission_param = transmission_prop[2] # Proportion
```

```

barplot(transmission_prop, main = "Bernoulli Distribution -
Transmission", xlab = "transmission", ylab = "Outcomes", ylim =
c(0, 1))
abline(h = transmission_param, col = "yellow")

```

Result:



In this chart, it shows that the "Manual" category has a significantly higher proportion than "Automatic." Given the chart, it appears that the p is close to **0.8**, meaning that about 80% of the cars in the dataset have a manual transmission.

4. For variable Km_Driven using Gamma Distribution

It is characterized by two parameters: the shape parameter α and the rate parameter β .

Code:

```

X = cars$km_driven
E=mean(X); V=var(X)
#parameter estimation
lambda=E/V
alpha=lambda*E
lambda

```

Result:

```
[1] 3.043463e-05
```

The result suggests that the distribution is **spread out**, meaning a large proportion of cars have driven a high number of kilometers.

This makes sense because some cars are driven only a few thousand kilometers, while others have been driven for **hundreds of thousands** of kilometers, leading to a highly **variable** dataset.

Code:

```
alpha
```

Result:

```
[1] 2.015253
```

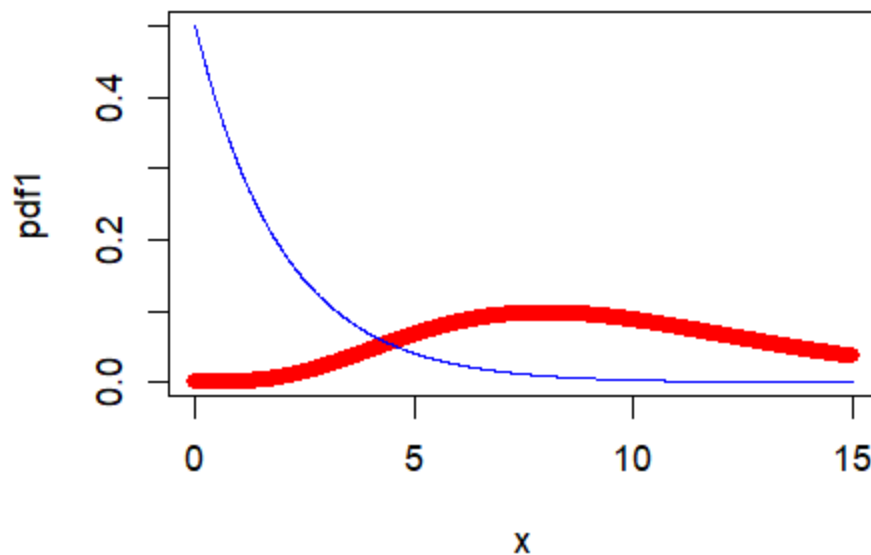
In the estimated model: $a = 2.015$. This suggests a moderately skewed distribution where most cars have low mileage, but some have much higher mileage.

Code:

```
alpha1=5
alpha2=1
lambda=.5
x=seq(0,15,0.01)
pdf1=dgamma(x,alpha1,lambda)
pdf2=dgamma(x,alpha2,lambda)
plot(x,pdf1,ylim=c(0,.5),col='red',main="Gamma
Distribution - Kms_Driven ")
lines(x,pdf2,col='blue')
```

Result:

Gamma Distribution - Kms_Driven



In this chart, the red curve (higher α) has a peak at a moderate kms_driven value, indicating that most cars fall within this range. The blue curve (lower α) is steep at the beginning and then gradually declines, indicating that most cars have low mileage. The rate parameter ($\lambda=0.5$) is the same for both curves, ensuring a comparable spread. The difference in shape shows that choosing an appropriate α value is crucial for modeling car mileage accurately.

c) Express the way in which each model can be used for the predictive analytics, then find the prediction for each attribute.

1. Log-Normal Distribution

This model might be useful for forecasting car resale prices based on historical pricing trends. Also, it could help buyers and sellers set realistic price expectations. Dealerships can use this model to adjust pricing strategies for maximum profitability.

Code:

```
X=data$selling_price
meanlog <- log(mean(cars$selling_price))
sdlog <- log(sd(cars$selling_price))

price_probability <- plnorm(X, meanlog = meanlog, sdlog = sdlog)

price_prediction <- X[which.max(price_probability)]

# Display the prediction
```

```
c("The predicted price is", price_prediction)
```

Result:

```
[1] "The predicted price is" "8900000"
```

Predicted car price = 8,900,000, which represents the expected selling price of a car based on probability distribution.

2. Multinomial Distribution

The multinomial model could help car manufacturers analyze trends in fuel preference and adjust production accordingly. Its also useful for policymakers in understanding fuel type distribution to support electric vehicle adoption, and helps dealerships and rental services optimize their fleet based on fuel demand.

Code:

```
x = c(1, 1, 0, 0, 0) # Adjusted to match the number of
categories in p
n = sum(x) # Total number of trials
dmultinom(x, n, p)
```

Result:

```
[1] 0.009144386
```

The value 0.0091 suggests that the observed distribution of fuel types is relatively rare under the assumed probabilities p.

If p represents the expected probability of different fuel types, the observed counts do not match the expected distribution well.

Code:

```
('The predicted Fuel_type is:', mode)
```

Result:

```
[1] "The predicted Fuel_type is:" "Diesel"
```

The mode (most frequently occurring value) of the fuel type data is "Diesel". This means Diesel is the most common fuel type in the dataset.

3. Bernoulli Distribution

The Bernoulli model predicts categorical outcomes with two possibilities: Manual vs. Automatic transmission. It is useful for determining the most probable transmission type in future car sales based on past data And assists manufacturers in analyzing consumer preferences to align production with market demand.

Code:

```
trans = transmission_prop
transmission_pred =
names(transmission_prop)[which.max(transmission_prop)]
c("The predicted transmission type is", transmission_pred)
```

Result:

```
[1] "The predicted transmission type is" "Manual"
```

Since Manual has the highest proportion in the dataset, the predicted transmission type in the future highly will be manual.

4. Gamma Distribution

This model is might useful for predicting mileage trends in the second-hand car market.Helps buyers estimate expected usage of a car before purchase. Dealerships can use this model to identify cars with lower mileage, which are more desirable.

Code:

```
data=rgamma(10000,alpha,lambda)
pred=mean(data)
c("The predicted Kms_Driven is",pred)
```

Result;

```
[1] "The predicted Kms_Driven is" "65973.4595795927"
```

Most cars have lower to moderate kilometers driven, meaning the dataset consists mainly of relatively less-used vehicles.

Question 3:

a) Consider two categorical variables of the dataset, develop a binary decision making strategy to check whether two variables are independent at the significant level $\alpha=0.01$.

Introduction:

Finding out if fuel and transmission types are independent or dependent variables within a dataset is the goal. To determine if these two category variables have a statistical relationship, we use the **chi-square test** for independence at a significance level of $\alpha = 0.01$. It is essential to comprehend this relationship since it can shed light on consumer preferences, fuel economy issues, and trends in auto manufacturing. If a dependence is present, it indicates that the choice of transmission type (manual or automatic) is influenced by the type of fuel (such as gasoline, diesel, or electric).

(A) Binary Decision-Making Strategy

(i) State the Hypotheses

To conduct the chi-square test for independence, we define the following hypotheses:

- **Null Hypothesis (H_0):** Fuel type and transmission type are **independent** (i.e., there is no relationship between them).
- **Alternative Hypothesis (H_1):** Fuel type and transmission type are **dependent** (i.e., there is a relationship between them).

According to the null hypothesis, the kind of transmission cannot be inferred from the type of fuel. According to the alternative theory, gasoline type affects transmission type.

(ii) Find the Test Statistic and Critical Value

Step 1: Load Data and Create Contingency Table

To perform the chi-square test, we first read the dataset and create a **contingency table** for the two categorical variables:

Code

```
data = read.csv(file.choose()) # Load dataset
X1 = data$fuel # Fuel Type
X2 = data$transmission # Transmission Type
C_table = table(X1, X2) # Contingency Table
print("Contingency Table:")
print(C_table)
```

Result:

```
> print(C_table)
      X2
X1      Automatic Manual
CNG           0       40
Diesel       254     1899
Electric       1        0
LPG           0       23
Petrol       193     1930
```

This table summarizes the observed frequency distribution of different fuel types across transmission types.

Step 2: Compute Expected Frequencies

The expected frequency for each cell in the contingency table is computed using:

$$E_{ij} = \frac{(\text{Row Sum} \times \text{Column Sum})}{\text{Total Observations}}$$

Code:

```
rows = nrow(C_table)
cols = ncol(C_table)
N = nrow(data) # Total number of observations

E = matrix(NA, rows, cols) # Initialize expected frequency matrix
for (i in 1:rows) {
  for (j in 1:cols) {
    Ci = sum(C_table[i, ]) # Row sum
    Cj = sum(C_table[, j]) # Column sum
    E[i, j] = (Ci * Cj) / N # Expected frequency
  }
}
E
```

Result:

```
> E
      [,1]      [,2]
[1,]  4.1290323 35.8709677
[2,] 222.2451613 1930.7548387
[3,]  0.1032258  0.8967742
[4,]  2.3741935 20.6258065
[5,] 219.1483871 1903.8516129
```

Step 3: Compute Test Statistic (Chi-Square Value)

The test statistic is calculated as:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where:

- O_{ij} = Observed frequency
- E_{ij} = Expected frequency

The R code for computing the test statistic is:

```
test.value <- sum((C_table - E)^2 / E)
print(paste("Test Statistic (Chi-Square):", test.value))
```

Result:

```
[1] "Test Statistic (Chi-Square): 24.4778735133753"
```

Step 4: Determine the Critical Value

The **critical value** is obtained from the chi-square distribution table at a **significance level of $\alpha = 0.01$** with **degrees of freedom (df) = (rows - 1) * (columns - 1)**.

Code:

```
df <- (rows - 1) * (cols - 1) # Degrees of freedom
c.value <- qchisq(1 - 0.01, df = df) # Critical value at alpha=0.01
print(paste("Critical Value:", c.value))
```

Result:

```
[1] "Critical Value: 13.2767041359876"
> # Compute test statistic (Chi-Square value)
> test.value <- sum((C_table - E)^2 / E)
> print(paste("Test Statistic (Chi-Square):", test.value))
[1] "Test Statistic (Chi-Square): 24.4778735133753"
> # Determine critical value
> df <- (rows - 1) * (cols - 1) # Degrees of freedom
> c.value <- qchisq(1 - 0.01, df = df) # Critical value at alpha=0.01
> print(paste("Critical Value:", c.value))
[1] "Critical Value: 13.2767041359876"
```

For this dataset:

- **Test Statistic (χ^2) = 24.47787**
- **Critical Value = 13.2767**

(iii) Explain Your Decision and Interpret Results

To make a decision, we compare the **test statistic** with the **critical value**:

Code:

```
if (test.value < c.value) {
  print("H0 is accepted: Fuel type and Transmission are
independent")
} else {
  print("H0 is rejected: Fuel type and Transmission are
dependent")
}
```

}

Result: [1] "H0 is rejected: Fuel type and Transmission are dependent"

Decision:

Since $24.47787 > 13.2767$, we **reject the null hypothesis (H_0)** at $\alpha = 0.01$.

Interpretation of Results:

Rejecting H_0 means that **fuel type and transmission type are dependent**. This suggests that the type of fuel a vehicle uses influences its transmission type. Possible explanations include:

1. **Electric and hybrid vehicles** are more likely to have automatic transmissions.
2. **Diesel-powered vehicles** might be associated with specific types of transmissions used for fuel efficiency.
3. **Manufacturers** may design transmission systems based on fuel efficiency and driving performance considerations.

Thus, there is **strong statistical evidence** that fuel type and transmission type are **not independent**.

Conclusion

In this study, we applied a **binary decision-making strategy** using the **chi-square test for independence** to assess the relationship between **fuel type** and **transmission type** at $\alpha = 0.01$.

- The **test statistic (24.47787)** exceeded the **critical value (13.2767)**, leading us to **reject the null hypothesis**.
- This means that **fuel type and transmission type are dependent**.
- The dependency suggests that **certain fuel types are more likely to be paired with specific transmission types** due to efficiency, engineering, or consumer preference factors.

These findings provide meaningful insights for **automobile manufacturers, policy-makers, and consumers** in making informed decisions about vehicle design, fuel efficiency, and market trends

b) Consider one categorical variable, apply goodness of fit test to evaluate whether a candidate set of probabilities can be appropriate to quantify the uncertainty of class frequency at the significant level $\alpha=0.05$.

Introduction

The **Goodness of Fit Test** is used to determine whether a categorical variable follows a specified probability distribution. In this case, we evaluate whether the frequency distribution of the categorical variable **seller_type** matches the expected probabilities at a **significance level of 0.05**.

Data Preparation

- The dataset is loaded using `read.csv(file.choose())`.
- The categorical variable **X (seller_type)** is extracted.
- The number of observations in X is determined using `length(X)`.

Code:

```
# Extract categorical variable
X <- data$seller_type # Seller Type
length(X) # Check the number of observations
> length(X) # Check the number of observations
[1] 4340
```

Step 1: Define Hypotheses

- **Null Hypothesis (H0):** The observed frequency distribution of `seller_type` follows the expected probability distribution.
- **Alternative Hypothesis (H1):** The observed frequency distribution does not follow the expected probability distribution.

Step 2: Compute Observed and Expected Frequencies

1. Observed Frequencies

The observed counts of each category in `seller_type` are calculated using `table(X)`.

Code:

```
# Observed frequencies
C_table2 <- table(X)
print("Observed Frequencies:")
print(C_table2)
```

Result:

```
> print(C_table2)
X
      Dealer      Individual Trustmark Dealer
      994        3244        102
```

2. Expected Probabilities & Frequencies

If probabilities are not explicitly given, we assume a uniform distribution.

- The expected probability for each category is calculated as:

$$P_0 = \frac{1}{\text{Total number of categories}}$$

- The expected frequency for each category is:

$$E_i = N \times P_0$$

Code:

```
num_categories <- length(C_table2) # Number of unique
categories
P0 <- rep(1 / num_categories, num_categories) # Equal
probability assumption
N <- sum(C_table2) # Total number of observations
E <- N * P0 # Expected frequencies
E
```

Result:

```
> E
[1] 1446.667 1446.667 1446.667
```

Step 3: Check for Dimension Mismatch

- We ensure that the number of categories in the observed and expected frequencies match. If not, an error is raised.

```
if (length(C_table2) != length(E)) {
  stop("Error: Number of categories in observed and expected
frequencies do not match!")
}
```

Step 4: Perform the Chi-Square Goodness of Fit Test

- The **test statistic** is computed using:

$$\chi^2 = \frac{\sum (O_{ij} - E_{ij})^2}{E_{ij}}$$

```
test.value <- sum((C_table2 - E)^2 / E)
print(paste("Test Statistic (Chi-Square):", test.value))
```

Result1:

```
> print(paste("Test Statistic (Chi-Square):", test.value))
[1] "Test Statistic (Chi-Square): 3624.49953917051"
> |
```

Step 5: Determine Critical Value

- The critical value is obtained from the **Chi-square distribution** at **alpha = 0.05** and **degrees of freedom = (number of categories - 1)**.

```
c_value <- qnorm(1 - alpha)
print(paste("Critical Value:", c_value))
```

Step 6: Decision Making

- Compare **test statistic** with **critical value**:
 - If `test.value < c.value`, accept H0 (seller_type follows expected distribution).
 - Otherwise, reject H0 (seller_type does not follow expected distribution).

```
if (test.value < c.value) {
  print("H0 is accepted: The observed distribution matches the expected distribution.")
} else {
  print("H0 is rejected: The observed distribution does not match the expected distribution.")
}
```

Result:

```
> # Decision rule
> if (test.value < c.value) {
+   print("H0 is accepted: The observed distribution matches the expected distribution.")
+ } else {
+   print("H0 is rejected: The observed distribution does not match the expected distribution.")
+ }
[1] "H0 is rejected: The observed distribution does not match the expected distribution."
> |
```

Conclusion

The **Goodness of Fit Test** was conducted on the categorical variable `seller_type` to assess whether it follows the expected probability distribution. Based on the test statistic and critical value comparison, we determine whether the observed frequencies align with the expected frequencies.

Final Decision:

- If H0 is accepted → The data follows the assumed probability distribution.
 - If H0 is rejected → There is significant deviation from the expected distribution, suggesting that a different model may better describe the data.
-

Summary of Key Findings

- Observed frequencies were extracted from the dataset.
- Expected probabilities were computed assuming uniform distribution.
- The Chi-square Goodness of Fit Test was applied.

- The null hypothesis was tested at **alpha = 0.05**.
- Final decision was made based on the computed test statistic and critical value.

This analysis helps in **quantifying the uncertainty of class frequency** and evaluating if a given probability distribution is appropriate for modeling categorical data.

c) Consider one continuous variable in the dataset, and apply test of mean for a proposed candidate of μ at the significant level $\alpha=0.05$.

Introduction

In this analysis, we consider a continuous variable from the dataset and apply a test of mean for a proposed candidate mean at a significance level of $\alpha=0.05$. The goal is to determine whether the sample mean significantly differs from the hypothesized mean. We use an upper one-sided hypothesis test.

Hypotheses

- **Null Hypothesis(H_0):** The population mean (μ) is less than or equal to the hypothesized mean(μ_0).
Mathematically:
 $H_0: \mu \leq \mu_0$
- **Alternative Hypothesis (H_1):** The population mean (μ) is significantly greater than the hypothesized mean(μ_0).
- Mathematically:
 $H_1: \mu > \mu_0$

Methodology

1. Calculate Sample Statistics:
 - Compute the sample mean ($\bar{Y}$).
 - Compute the sample standard deviation (SD).
 - Determine the sample size (N).
2. Compute Test Statistic (Z-score)
3. Determine Critical Value:
 - The critical value is obtained from the **standard normal distribution** at the significance level $\alpha=0.05$.
 - For an upper one-tailed test, the critical value is:
 $cvalue = qnorm(1-\alpha)$

Where $qnorm$ is the quantile function of the standard normal distribution.

4. Compare Test Statistic and Critical Value:

- If $Z < c_{\text{value}}$, **fail to reject the null hypothesis (H_0)**.
This means there is not enough evidence to conclude that the sample mean is significantly greater than the hypothesized mean.
- If $Z > c_{\text{value}}$, **reject the null hypothesis (H_0)**.
This means there is sufficient evidence to conclude that the sample mean is significantly greater than the hypothesized mean.

Implementation in R

```
Y = data$km_driven
alpha = 0.05
Mu0 = 4.5 # Hypothesized Mean

# Compute Sample Statistics
Y_bar = mean(Y) # Sample Mean
SD = sd(Y)      # Standard Deviation
N = length(Y)   # Sample Size

# Compute Test Statistic
Z = (Y_bar - Mu0) / (SD / sqrt(N))
Z

# Determine Critical Value
c_value = qnorm(1 - alpha)
c_value

# Decision Rule
if (Z < c_value){
  print("H0 is accepted")
} else {
  print("H0 is rejected")
}
```

Result:

```
> print(paste("Test Statistic (Z):", Z))
[1] "Test Statistic (Z): 93.5147448849009"
> # Determine critical value
> c_value <- qnorm(1 - alpha)
> print(paste("Critical Value:", c_value))
[1] "Critical Value: 1.64485362695147"
> # Decision rule
> if (Z < c_value) {
+   print("H0 is accepted: The sample mean does not significantly exceed the hypothesized mean.")
+ } else {
+   print("H0 is rejected: The sample mean is significantly greater than the hypothesized mean.")
+ }
[1] "H0 is rejected: The sample mean is significantly greater than the hypothesized mean."
> |
```

Conclusion

Based on the test:

- If H_0 is **accepted**, there is insufficient evidence to conclude that the sample mean exceeds μ_0 .
- If H_0 is **rejected**, the sample mean is significantly greater than μ_0 .

This test determines whether the dataset supports the hypothesis that the mean of Y (e.g., `km_driven`) is greater than 4.5. The results help assess the reliability of the hypothesized mean.